TRIBHUVAN UNIVERSITY

**Saraswati Multiple Campus**

Lekhnath Marga, Thamel

Preboard 2025 May 26, 2025

Bachelor in Computer Application

Course Title: OOP in Java
Code No
Semester: 3rd

Full Marks 60
pass Marks 24
Time: 3 hours

## Group B

Answer any Six 6X5=30

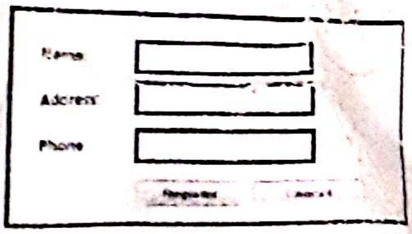
2. Explain the working of JVM in detail? Explain any 3 string methods [2.5X2=5]
3. Explain different operator in Java. Write a program to find the factorial of a program using constructor. WAP to find the [1+2+2] → random number
4. What is Exception handling? Write a program using "NullPointerException" and 'IO Exception' with example [1+2+7]
5. Discuss the various levels of access protection available for packages and their implications.[5]
6. Write short Notes on [2.5+2.5]
   a. Life cycle of Thread
   b. "this" keyword
7. How user defined Packages are created? Write a program that create a package to add two **Integer number?** **[2+3]**
8. What is Layout management? Write a program to display four Buttons ***"Add, Subtract, Multiply, Divide"*** using "***grid layout***". [1+4]

## Group C

Attempt any TWO Questions

9.
   a. Write a Program to show how a Thread is created by implementing Runnable interface.[5]
   b. Write a program to create the Form using Java Code., which have two ***label***(First num, Second Number), three ***Text Fields***(First Number, Second Number) , three ***Button*** (Divide and Multiply Cancel). [5]

10.
   a. What is Applet? Write a program to display circle and rectangle using Applet.? [1+4]
   b. For the following Form , save the data in the database when " ***Register"*** Button is clicked. (DataBase:"BCA", Table:"Student (sname, saddress, sphone)




11.
   a. Write a program to sort an Array in Java using Bubble Sort method.WAP to create a file and wite "hello world " in bca.txt in java [2.5X2=5]
   b. What is the difference between Method Overloading and Method Overriding ,explain with examples[2.5+2.5]

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Tribhuvan University

**Saraswati Multiple Campus**

Lekhnath Marga, Thamel

Preboard,2082(2025 05 30), **Probability and Statistics**

Any Six (6X5=30)

**Attempt any six questions** **Group "B"**

**[6*5=30]**

**2.** Describe the characteristics of binomial and Poisson distributions.

3. The probability that Nepal wins a football match against Bangladesh is 2/3. If Nepal and Bangladesh play three matches, what is the probability that:

(i) Will Nepal lose all three matches?

(ii) Will Nepal win at least one match?

4. Following are the marks obtained by two students, A and B, in 10 tests of 100 marks each:

| Tests | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Marks secured by A | 44 | 80 | 76 | 48 | 52 | 72 | 68 | 56 | 60 | 54 |
| Marks Secured by B | 48 | 75 | 54 | 60 | 63 | 69 | 72 | 51 | 57 | 66 |

If the consistency of performance is the criterion for awarding a prize, who should get the prize?

5 If the correlation coefficient of a sample of 9 pairs of observations is 0.5, find the standard error and probable error. Also, determine the limits of the population correlation coefficient.

6 On average, one in 400 items is defective. If items are packed in boxes of 100, what is the probability that any given box of items will contain (i) no defectives, (ii) fewer than two defectives, (iii) one or more defectives, (iv) more than three defectives

7. The following table shows the number of motor registrations and the sales of Gorakhali Motor tires by a wholesale dealer in Kathmandu over a 5-year period.

| **Year** | **Motor Registration ('000' Nos.)** | **No. of tyres sold ('000' Nos)** |
|---|---|---|
| 1 | 60 | 125 |
| 2 | 63 | 110 |
| 3 | 72 | 130 |
| 4 | 75 | 135 |
| 5 | 80 | 150 |

(i) Estimate the sales of tires when the expected motor registration in the next year is 90,000.

(ii) Also, calculate the correlation coefficient and interpret.

8. Applicants for a certain job are given an aptitude test. Past experience shows that the scores from the test are normally distributed with a mean of 60 points and a standard deviation of 12 points. What percent of candidates would be expected to pass the test if a minimum score of 75 is required?

**Group "C"**

**Attempt any two questions** **10*2=20**

9 a) A project yields an average cash-flow of Rs. 500 lakhs with a standard deviation of Rs. 60 lakhs. Calculate the following probabilities.

(i) Cash flow will be more than Rs. 560 lakhs.

(ii) Cash flow will be less than Rs. 420 lakhs.

(iii) Cash flow will be between Rs. 460 lakhs and Rs. 540 lakhs.

(iv) Cash flow will be more than Rs. 680 lakhs.

b) Write down any five properties of the Normal distribution

10. The following are the defective pieces produced by four operators working, in turn, on four different machines.

| Machine | Operators B/1 | B/2 | B/3 | B/4 |
|---|---|---|---|---|
| A/1 | 34 | 28 | 33 | 29 |
| A/2 | 31 | 24 | 35 | 23 |
| A/3 | 27 | 20 | 43 | 72 |
| A/4 | 28 | 28 | 29 | 26 |

Perform analysis of variance at the 0.05 level of significance to ascertain whether variability in production is due to variability in operators' performance or variability in machines' performance.

11. A sample of polythene bags from two manufacturers, A and B, is tested by a prospective buyer for bursting pressure, and the results are as follows:

| **Bursting Pressure (lb.)** | **No. of Bags** A | B |
|---|---|---|
| 5.0 – 9.9 | 2 | 9 |
| 10.0 – 14.9 | 9 | 11 |
| 15.0 – 19.9 | 29 | 18 |
| 20.0 – 24.9 | 54 | 32 |
| 25.0 – 29.9 | 11 | 27 |
| 30.0 – 34.9 | 5 | 13 |

Which set of bags has more uniform pressure? If prices are the same, which manufacturer's bags would be preferred by the buyer? Why?

"BEST OF LUCK"

# TRIBHUVAN UNIVERSITY

Faculty of Humanities & Social Science

SARASWATI MULTIPLE CAMPUS,

Preboard 2025, 5/28/2025

**B. Attempt any six questions 6X5=30**

1. What is CSS selector? Explain different types of CSS selectors with suitable example.
2. Write HTML tag to generate following table?

<table>
<tr><td colspan="4">Routine</td></tr>
<tr><td>Sunday</td><td>Monday</td><td>Break</td><td>Tuesday</td></tr>
<tr><td>WT</td><td>DSA</td><td rowspan="3"></td><td>SAD</td></tr>
<tr><td>Java</td><td>Statistics</td><td>Maths</td></tr>
<tr><td></td><td></td><td></td></tr>
</table>

3. What is exception handling? Explain with suitable example?
4. What is XML schema? How to define attributes and elements into xml schema? xs: attr
5. What is XSL? Describe XSLT with example?
6. What is cookie? Describe the cookie with example?
7. What is SQL statement? Describe the different SQL statement with example?
8. Write short notes on:
   a. Xquery b.Xparser

**C. Attempt any two questions 2X10=20**

9. Different between tags and attributes? Write a HTML and CSS code to design your curriculum vitae (CV) using different HTML elements like ( table, image, formatting tags, links).
10. Create an attractive form in asp dot net and display form data in another page.
11. What is database connectivity? How can you configure database connection to SQL server database? Write a query to insert data into the database with the help of ASP.Net?

*******Best of Luck******

Tribhuvan University
Faculty of Humanites and Social Sciences
**Saraswati Multiple Campus**
082/2/18(PREBOARD )

Bachelors in Computer Applications
Course: System Analysis and Design
Code No:
Semester : 3rd Sem

Full Marks:60
Pass Mark:24
Time :3 Hrs
Batch: SAGARMATHA (2078)

Group B

Attempt all the questions. (5 X 6 =30)

1. Explain Prototype system development model with advantages and disadvantages.

2. Create a PERT chart for a project with the following tasks and durations:

- Task A: 5 days
- Task B: 4 days
- Task C: 3 days
- Task D: 2 days
- Task E: 6 days
- Task F: 3 days
- Task G: 4 days
- Dependencies: A → B, A → C, B → D, C → E, D → F, E → G, F → G

Identify the critical path and explain its significance for the project schedule.

3. Explain system Maintenance with its type.

4. Given the following table for a student registration system:

Perform normalization up to the Third Normal Form (3NF) and provide the resulting tables.

| Sid | Stdname | Cid | Cname, | Instructor | InstructorPhone |
|---|---|---|---|---|---|
| 001 | Ram | C101 | Java | Srawan | 1234 |
| 001 | Ram | C102 | Python | Dipak | 5678 |
| 001 2 | Hari | C101 | Java | Srawan | 1234 |

5. A company wants to develop an inventory management system. Identify and describe three key functional requirements and two non-functional requirements for this system. Provide a brief explanation for each requirement.

6. What are the primary types of "testing" performed during the software development lifecycle?

## Group C

(10 X 2=20)

**Attempt all the questions.**

1. A tech company is developing a new customer relationship management (CRM) software. The following financial details are provided:

**- One-Time Initial Costs:**

- Software Development: NRS 60,00,000
- Infrastructure Setup: NRS 9,00,000
- Initial Advertising: NRS 6,00,000

**- Annual Recurring Costs:**

- Technical Support: NRS 10,00,000
- Licensing Fees: NRS 5,00,000
- Operational Costs: NRS 3,00,000

**- Annual Revenue Projections:**

- Year 1: NRS 75,00,000
- Year 2: NRS 95,00,000
- Year 3: NRS 1,20,00,000
- Year 4: NRS 1,40,00,000
- Year 5: NRS 1,60,00,000
- Discount Rate:9%

a. Calculate the Total Initial Costs for the software project.

b. Determine the Total Annual Costs for each year, including both recurring costs and the annual portion of the initial costs (amortized over 5 years).

c. Calculate the Break-Even Point in Dollars for each year based on the annual revenue projections.

d. Perform a Net Present Value (NPV) Calculation of the projected net cash flows over 5 years, considering the discount rate.

e. Discuss the implications of the break-even analysis and NPV calculation on the company's investment and financial planning.

2. Design a Level 0 and Level 1 Data Flow Diagram for a Customer Support System. Include key processes, data stores, and external entities.
3. Design an E-R diagram for a Retail Inventory Management System. Include entities such as Product, Supplier, Order, and Customer, and define their relationships and attributes. Describe how the E-R diagram helps in designing and managing the database effectively.

Tribhuvan University

**Saraswati Multiple Campus**

Thamel , Kathmandu

PREBOARD, 2025/5/27

**Semester:3rd , Annapurna/Machhapuchre** **Data Structure and Algorithm**

**Batch:080 Batch**

Group:B

Attempt any 6 question. [6x5=30]

1. What is data structure .Explain about divide and conquer strategy with example?
2. What is collision in hashing? How collision resolution can be done using separate chaining method? Explain
3. List any to application of Stack? Convert A+(B*C-(D/E-F)*G)*H into postfix expression.
4. Write an algorithm and a program to implement basic Queue Operation?
5. What is AVL tree? Construct AVl tree of 50, 40, 35, 58, 48, 42, 60, 30, 33, 32
6. What is double linked list? Write an algorithm to add a node at the beginning and end of double linked list?
7. Define graph. Explain breadth first traversal and depth first traversal with example.

Group C

Answer any two questions[2X10=20]

8. What is sorting .Differentiate between internal and external sorting .write a function to implement heap sort and sort the following using heap sort:12,9,1,13,16,24,215
9. What is stack? List the application of stack. Write an algorithm to perform PUSH and POP operation?
10. Explain different types of graph. Explain prims algorithm with example